Project Progress Report

Amity School of Engineering & Technology

Project Group ID - 265

Project Area – Data Science (Machine Learning)

B.Tech (8CSE-DS-1Y) VIII Semester

Project Progress Report

Students Project Progress Report for Even Semester of session 2025-2026

Project Title: Machine Learning based Predictive Model for Livability using Environmental and Socioeconomic Indicators

Project Group: 265

Students:

1. Sagar Singh Bisht
   Roll No. 056,
   Enrollment No. A023167022056

2. Aakarsh Pab
   Roll No. 050,
   Enrollment No. A023167022050

1. Synopsis Submission

- The project synopsis was revisited and pivoted from a generic "Predictive Model" to a specialized **"Recommender System" (CityMatch)**. The new objective—solving the multi-criteria optimization problem for urban relocation using Clustering and Vector Space Matching—was defined and approved.

2. Literature Review

- A targeted literature review was conducted, shifting focus from regression analysis to **Unsupervised Learning** and **Decision Support Systems**.
- **Key Domains Explored:** K-Means Clustering for urban segmentation, Content-Based Filtering for recommendation logic, and Geospatial Analytics (OSMnx).
- **Key Outcome:** Identified "Recency Bias" in existing studies and resolved to use **Historical Time-Series Data** (2023-2025) rather than static snapshots.

3. Indicator Selection & Data Sourcing

- We expanded the scope from the initial 15 monitoring stations to a **mesh of 50+ residential neighborhoods**.
- **Environmental:** Historical hourly PM2.5/NO2 data (Jan 2023–Dec 2024) via **Open-Meteo API**.
- **Infrastructure:** Hyper-local amenity density (Schools, Hospitals, Parks, Metro) via **OpenStreetMap (OSMnx)**.
- **Socioeconomic:** Rental yield estimates and derived Traffic Indices based on pollutant proxies.

4. Data Collection & Preparation (Automated Pipelines)

- Moved from manual data entry to **Automated Python Pipelines**:
- **Script 1 (Geospatial):** Utilized OSMnx to scrape and count amenities within a 1km radius of all 50 target neighborhoods.
- **Script 2 (Temporal):** Fetched ~25,000 hourly data points per station to compute robust **Seasonal Profiles** (Winter vs. Monsoon averages), eliminating the "seasonality error" flagged in the minor project.

5. Exploratory Data Analysis (EDA)

- **Clustering Analysis:** Used the **Elbow Method** and **Silhouette Analysis** to determine the optimal number of neighborhood clusters ($k=4$).
- **Pattern Discovery:** Identified distinct neighborhood archetypes (e.g., "Affordable Industrial" vs. "Premium Green Zones") that were not visible in simple average plots.

6. Data Cleaning and Processing

- **Imputation:** Handled missing environmental data using **Spatial Interpolation** (Nearest Neighbor Mapping).
- **Normalization:** Applied **Min-Max Scaling** to normalize disparate units (Rent in ₹ vs. PM2.5 in $\mu g/m^3$) to the [0, 1] range.
- **Inversion Logic:** Implemented feature inversion ($1 - x$) for negative attributes (Pollution, Cost) to ensure they function correctly in similarity calculations.

7. Modeling Preparation & Implementation

- **Clustering Engine:** Successfully implemented **K-Means Clustering** to segment the city into 4 socioeconomic clusters.

- **Recommender Engine:** Built a **Cosine Similarity** logic layer that accepts a user's priority vector (e.g., [Clean Air: 10, Cheap Rent: 10]) and ranks neighborhoods by mathematical proximity.
- **Dashboarding:** Initiated development of a **Streamlit Web Interface** to visualize these recommendations interactively.

8. Challenges and Reflections

- **Data Scarcity:** Real estate data was difficult to scrape due to anti-bot protections. *Solution:* We conducted a manual domain survey for the 50 neighborhoods to build a "Ground Truth" rent column.
- **API Limits:** Sequential fetching of historical data was slow. *Solution:* Implemented batch processing with delays to respect API rate limits.

---

9. Summary of Project Progress (WPR Timeline)

WPR	Week / Dates	Key Activities & Progress
1	Week 1	Pivot & Design: Redefined problem statement to "CityMatch"; defined architecture (Clustering + Recommender).
2	Week 2	Data Acquisition: Built OSMnx scraper for amenities and Open-Meteo fetcher for 2-year history.
3	Week 3	Preprocessing: Merged geospatial and temporal datasets; Manual Rent entry; Normalization pipeline.
4	Week 4	ML Implementation: Ran K-Means Clustering (k=4); Developed Cosine Similarity logic; Cluster profiling.

---

10. Supervisor's Assessment

- Supervisor has reviewed the shift to the Recommender System architecture and marked the progress in automated data acquisition and unsupervised learning implementation as satisfactory.

11. Conclusion & Forward Plan

- The project has successfully transitioned from a static analysis to a functional AI-driven Application.
- Current Status: The core ML engine (Clustering + Matching) is operational.
- Next Steps: Finalize the Streamlit Dashboard, conduct "Persona Testing" (verifying results for different user types), and write the final thesis report.

Student Signature:

Sagar Singh Bisht:

Aakarsh Pab:

Faculty Guide Signature: